

Software Requirements Specification

FoneBox Enterprise ICT Solutions Platform

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) provides a comprehensive description of the FoneBox Enterprise ICT Solutions Platform. It outlines the architectural design, functional and non-functional requirements, user interfaces, and the overarching security protocols governing the platform. This document serves as the definitive reference for stakeholders, development teams, and system auditors.

1.2 Scope

The FoneBox Enterprise ICT Solutions Platform is a highly secure, multi-tenant portal designed to manage state-level cryptographic operations, enterprise resource planning, and sovereign citizen identity ecosystems.

Key functional domains include:

- **Public Corporate Portal:** Client acquisition, security briefings, and demonstrations.
- **Client Management Portal:** Post-authentication dashboard for enterprise clients to manage billing, infrastructure, and encrypted support channels.
- **Operations & Security Command Center (SOC):** Internal tools for threat intelligence, audit logging, and cryptographic hardware management.
- **CRM & Sales Operations:** Internal lead management, invoicing, and sales pipeline tracking.
- **Super Administrator Console:** Root-level oversight of user access, system logs, and CMS settings.

1.3 Definitions and Acronyms

- **HSM:** Hardware Security Module
- **FIPS 140-3:** Federal Information Processing Standard for cryptographic modules
- **PKI:** Public Key Infrastructure
- **SOC:** Security Operations Center
- **CRM:** Customer Relationship Management
- **CMS:** Content Management System

2. Overall Description

2.1 System Architecture

The platform is engineered using a modern JavaScript/TypeScript stack

- **Frontend Layer:** React.js, Vite, Tailwind CSS, utilizing a mobile-first responsive design paradigm with hardware-accelerated animations (Framer Motion).
- **Backend Services:** Node.js with Express.js, strictly typed with TypeScript.
- **Database Layer:** PostgreSQL managed via Prisma ORM for type-safe database queries and migrations.
- **Security:** JWT-based authentication, FIDO2 WebAuthn readiness, and bcrypt password hashing.

2.2 User Characteristics and Roles

1. **SUPER_ADMIN:** Absolute control over all portals, user provisioning, system settings (CMS), and audit logs.
2. **CORPORATE_CLIENT:** Enterprise users accessing infrastructure health, billing, and secure communications.
3. **SALES_EXEC:** Internal sales personnel managing inbound leads, consultations, and PDF invoice generation.
4. **OPERATIONS_OFFICER:** Security analysts and operations engineers managing PKI infrastructure and threat intelligence.

3. Specific Functional Requirements

3.1 Public Corporate Portal

The public-facing website acts as the primary acquisition channel.

- **Hero & Solutions Showcases:** Interactive UI detailing E-Passports, Secure Bank Cards, and Electronic Visas.
- **Lead Generation:** Modals for "Request Consultation", "Request Product Demo", and "Security Audit Vetting".
- **Dynamic Content:** Newsletter subscription and Careers application systems. All submissions route directly to the backend CRM.

3.2 Authentication & Authorization

- **Login Modal:** Secure entry point supporting username/password and multi-factor authentication (MFA) paradigms.
- **Role-Based Access Control (RBAC):** Post-login routing strictly directs users to their designated operational portals.

3.3 Super Admin Portal

- **Dashboard:** Real-time metrics on global server load, cryptographic key usage, and system health.
- **User Management:** Full CRUD (Create, Read, Update, Delete) capabilities over system users.
- **CMS Management:** Ability to dynamically update social links and public site configurations.

3.4 Client Portal

- **Infrastructure Overview:** Visual representation of server statuses and active nodes.

- **Billing & Invoices:** View and download securely generated PDF invoices.
- **Support Command:** End-to-end encrypted ticketing system for incident reporting.

3.5 CRM & Sales Portal

- **Lead Pipeline:** Centralized dashboard to track inbound requests (Consultations, Demos, Audits).
- **Invoice Generation:** Tooling to automatically generate and dispatch professional PDF invoices to clients.

3.6 SOC Analyst Portal

- **Threat Intelligence:** Real-time logging of system anomalies and failed authentication attempts.
- **Audit Logs:** Immutable tracking of user actions across the platform for compliance.

4. External Interfaces

4.1 Application Programming Interfaces (APIs)

- **RESTful Endpoints:** The backend exposes `/api/crm`, `/api/cms`, and `/api/auth` endpoints.
- **Data Exchange:** All payload data is formatted in strictly defined JSON.

4.2 Database Schema Overview

The database employs relational modeling via Prisma:

- **User:** Manages authentication credentials and role assignments.
- **ConsultationRequest / Lead:** Centralizes inbound public requests.
- **SupportCase:** Tracks internal and client-facing tickets.

5. Non-Functional Requirements

5.1 Security & Compliance

- **Encryption:** All data in transit is secured via TLS 1.3. Passwords are salted and hashed using bcrypt.
- **FIPS Alignment:** The system architecture is designed to interface securely with FIPS 140-3 Level 4 certified hardware.
- **Data Sovereignty:** Database designs support localized hosting to comply with national data sovereignty laws.

5.2 Performance & Reliability

- **Responsive Design:** The UI is fully optimized for desktop and mobile devices via custom scrollbar utilities and fluid grid layouts.
- **Uptime:** The Node.js application is managed via PM2 process manager on the VPS, ensuring automatic restarts and high availability.

5.3 Maintainability

- **Code Structure:** Component-driven architecture using React ensures scalable UI expansion.
- **Automated Deployment:** Continuous deployment scripts (`fixVPS.ts`) automate git pulls, builds, and PM2 restarts without downtime.

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